

ES 215: Computer Organization and Architecture

A Review on Digital Systems and Design



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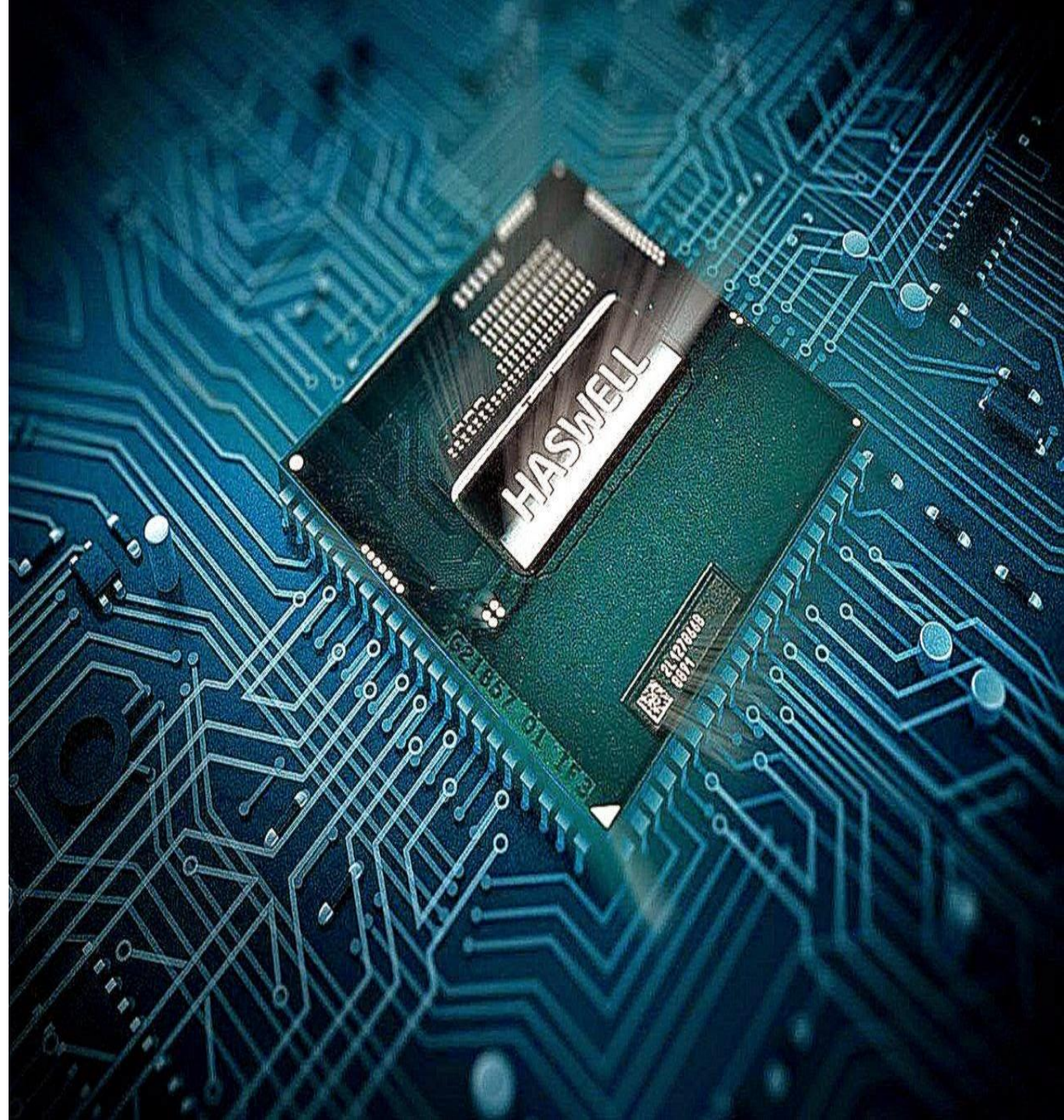
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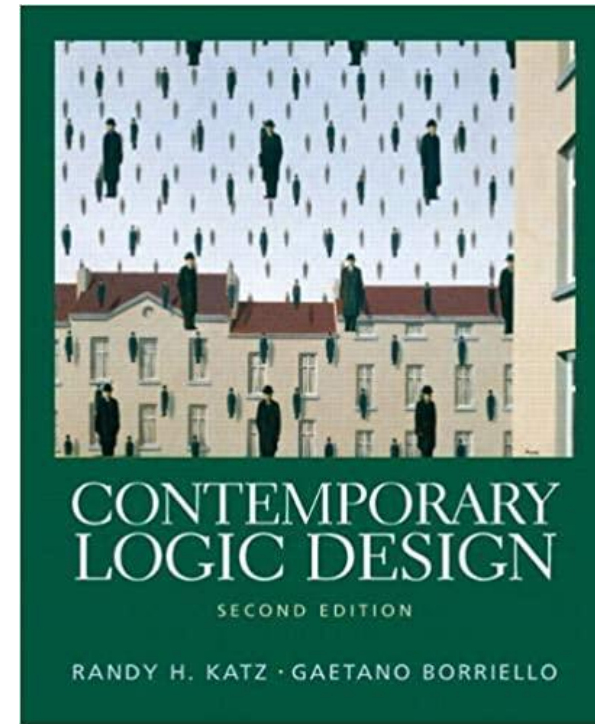
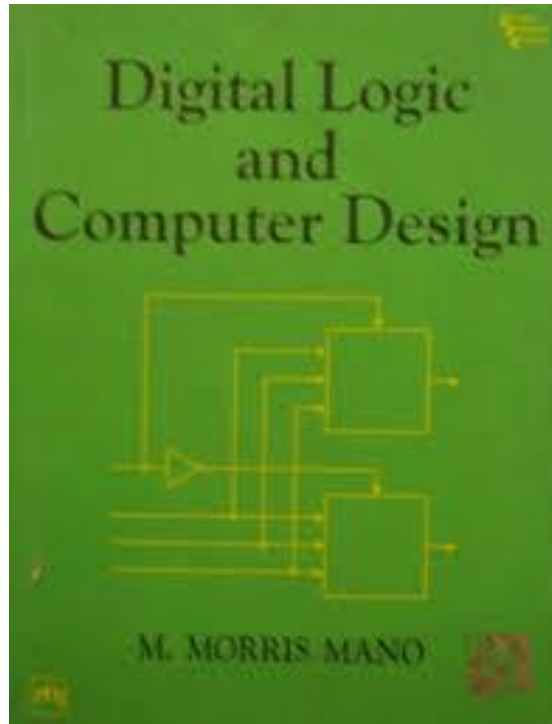
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Content

- Digital Systems & Design
 - Under the Hoods of COA
- Coverage
 - Basics of Digital Systems & Design Principles
 - Only the aspects relevant to the teachings of COA*.

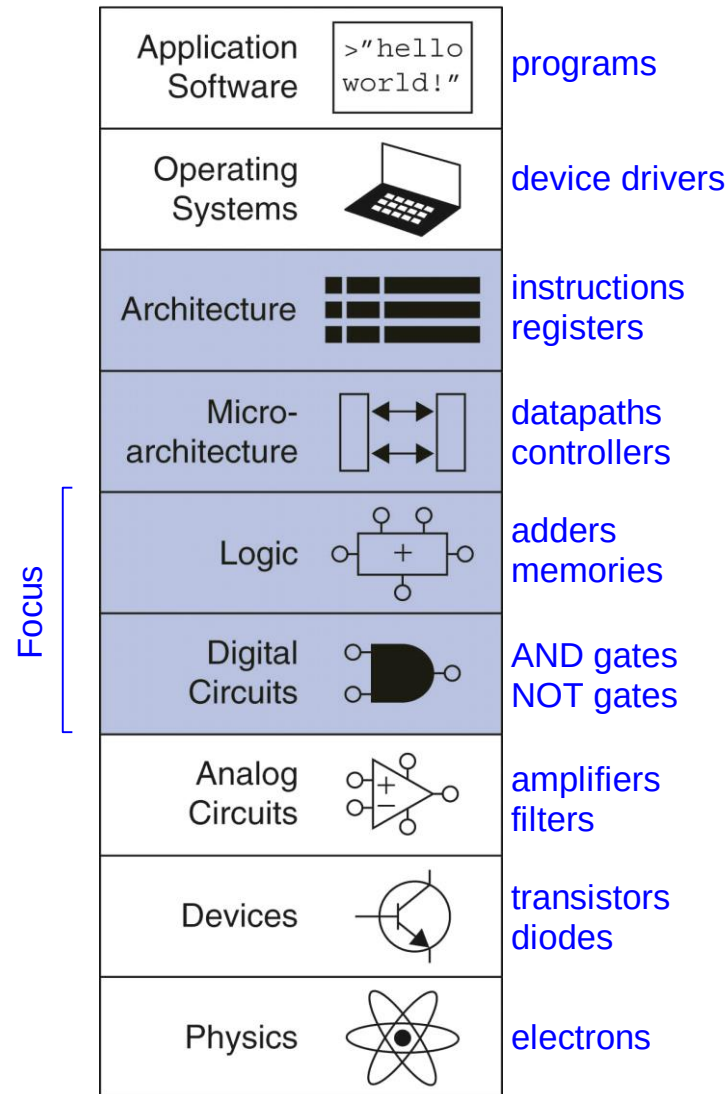


BOOKS AND MATERIALS REFERRED

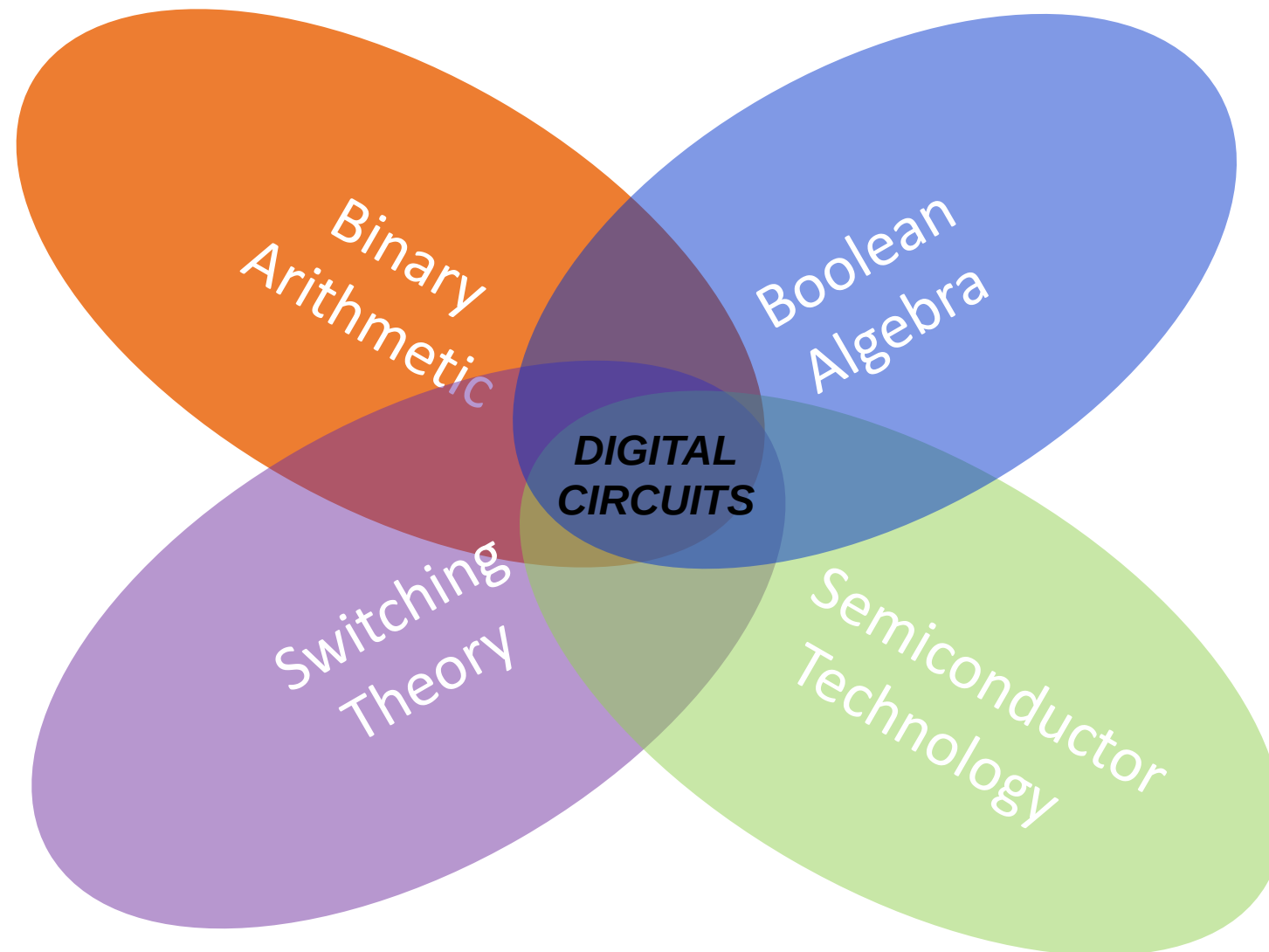


Abridged version of the presentation slides are adopted from the materials from the above books.

DIGITAL DESIGN – WHERE IT ALL FITS?



DIGITAL SYSTEMS (FUNDAMENTALS)

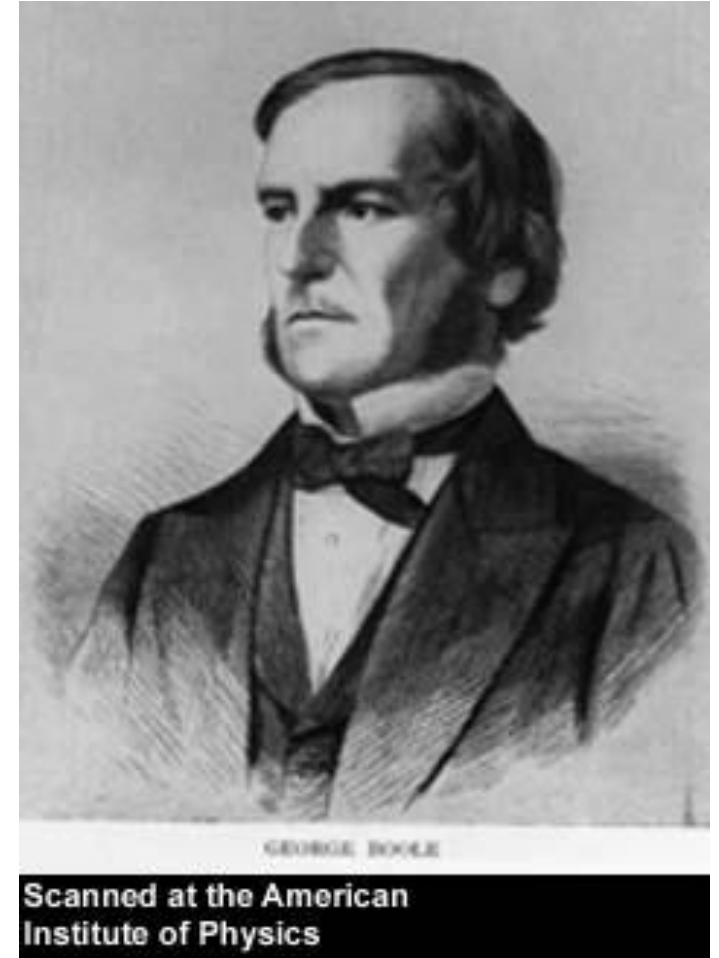


- Most physical variables are continuous
 - Voltage on a wire
 - Frequency of an oscillation
 - Digital abstraction considers discrete subset of values.
 - Intentionally restrict design choices
 - Examples: i.e., digital cameras, digital television, cell phones, CDs
 - Discrete voltages instead of continuous
 - Discrete (Digital) Clock
- *Simpler to design than analog circuits – can build more sophisticated systems.*
- *Digital systems replacing analog predecessors.*

- **Two discrete values:** 1's and 0's
 - 1, TRUE, HIGH
 - 0, FALSE, LOW
- **1 and 0:** voltage levels, clock signal, keypad input, etc.
- Digital circuits use **voltage** levels to represent 1 and 0
- ***Bit:*** Binary digit

GEORGE BOOLE, 1815-1864

- Born to working class parents
- Taught himself mathematics and joined the faculty of Queen's College in Ireland
- Wrote *An Investigation of the Laws of Thought* (1854)
- **Introduced binary variables**
- Introduced the three fundamental logic operations: AND, OR, and NOT



- Decimal numbers

1's column
10's column
100's column
1000's column

$$5374_{10} = 5 \times 10^3 + 3 \times 10^2 + 7 \times 10^1 + 4 \times 10^0$$

five three seven four
thousands hundreds tens ones

- Binary numbers

1's column
2's column
4's column
8's column

$$1101_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 13_{10}$$

one one no one
eight four two one

- Decimal to binary conversion:

- Convert 47_{10} to binary
- $32 \times 1 + 16 \times 0 + 8 \times 1 + 4 \times 1 + 2 \times 1 + 1 \times 1 = 101111_2$

- Binary to Decimal conversion:

- Convert 10011_2 to decimal
- $16 \times 1 + 8 \times 0 + 4 \times 0 + 2 \times 1 + 1 \times 1 = 19_{10}$

- N -digit decimal number
 - How many values?
 - Range?
 - Example: 3-digit decimal number:
- N -bit binary number
 - How many values?
 - Range:
 - Example: 3-digit binary number:

- N -digit decimal number
 - How many values? 10^N
 - Range? $[0, 10^N - 1]$
 - Example: 3-digit decimal number:
 - $10^3 = 1000$ possible values
 - Range: $[0, 999]$
- N -bit binary number
 - How many values? 2^N
 - Range: $[0, 2^N - 1]$
 - Example: 3-digit binary number:
 - $2^3 = 8$ possible values
 - Range: $[0, 7] = [000_2 \text{ to } 111_2]$

HEXADECIMAL NUMBERS

- Base 16
- Shorthand for binary

Hex Digit	Decimal Equivalent	Binary Equivalent
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
A	10	1010
B	11	1011
C	12	1100
D	13	1101
E	14	1110
F	15	1111

- Hexadecimal to binary conversion:
 - Convert $4AF_{16}$ (also written 0x4AF) to binary
- Hexadecimal to decimal conversion:
 - Convert 0x4AF to decimal

- Hexadecimal to binary conversion:
 - Convert $4AF_{16}$ (also written $0x4AF$) to binary
 - $0100\ 1010\ 1111_2$
- Hexadecimal to decimal conversion:
 - Convert $4AF_{16}$ to decimal
 - $16^2 \times 4 + 16^1 \times 10 + 16^0 \times 15 = 1199_{10}$

BITS, BYTES, NIBBLES...

- Bits

10010110

most significant bit least significant bit

- Bytes & Nibbles

byte

10010110

nibble

- Bytes

CEBF9AD7

most significant byte least significant byte

POWERS OF 2

- $2^0 = 1$
- $2^1 = 2$
- $2^2 = 4$
- $2^3 = 8$
- $2^4 = 16$
- $2^5 = 32$
- $2^6 = 64$
- $2^7 = 128$
- $2^8 = 256$
- $2^9 = 512$
- $2^{10} = 1024$
- $2^{11} = 2048$
- $2^{12} = 4096$
- $2^{13} = 8192$
- $2^{14} = 16384$
- $2^{15} = 32768$
- Handy to memorize up to 2^9

LARGE POWERS OF TWO

- $2^{10} = 1 \text{ kilo}$ ≈ 1000 (1024)
- $2^{20} = 1 \text{ mega}$ $\approx 1 \text{ million}$ (1,048,576)
- $2^{30} = 1 \text{ giga}$ $\approx 1 \text{ billion}$ (1,073,741,824)

- What is the value of 2^{24} ?
- How many values can a 32-bit variable represent?

- What is the value of 2^{24} ?

$$2^4 \times 2^{20} \approx \mathbf{16 \text{ million}}$$

- How many values can a 32-bit variable represent?

$$2^2 \times 2^{30} \approx \mathbf{4 \text{ billion}}$$

- Decimal

$$\begin{array}{r} 11 \leftarrow \text{carries} \\ 3734 \\ + 5168 \\ \hline 8902 \end{array}$$

- Binary

$$\begin{array}{r} 11 \leftarrow \text{carries} \\ 1011 \\ + 0011 \\ \hline 1110 \end{array}$$

- Digital systems operate on a **fixed number of bits**
- Overflow: when result is too big to fit in the available number of bits
- See previous example of $11 + 6$

- Sign/Magnitude Numbers
- Two's Complement Numbers

SIGNED/MAGNITUDE NUMBERS

- 1 sign bit, $N-1$ magnitude bits
- Sign bit is the most significant (left-most) bit
 - Positive number: sign bit = 0
 - Negative number: sign bit = 1

$$A: \{a_{N-1}, a_{N-2}, \dots, a_2, a_1, a_0\}$$

$$A = (-1)^{a_{n-1}} \sum_{i=0}^{n-2} a_i 2^i$$

- Example, 4-bit sign/mag representations of ± 6 :
 - +6 = **0110**
 - 6 = **1110**
- Range of an N -bit sign/magnitude number:
 $[-(2^{N-1}-1), 2^{N-1}-1]$

- Problems:
 - Addition doesn't work, for example $-6 + 6$:

$$\begin{array}{r} 1110 \\ + 0110 \\ \hline 10100 \text{ (wrong!)} \end{array}$$

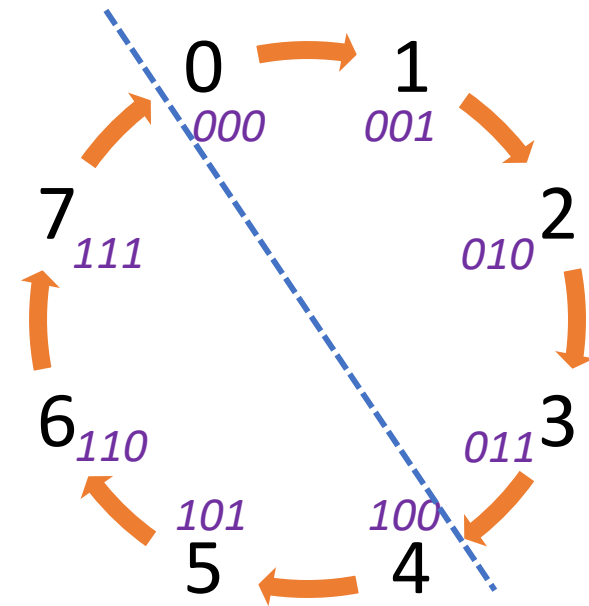
- Two representations of 0 (± 0):

1000

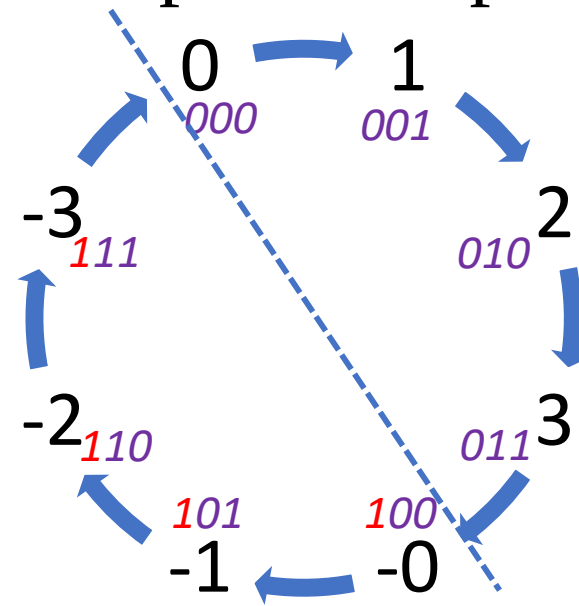
0000

NUMBER CYCLE: UNSIGNED, SIGNED SIGN MAGNITUDE, 1's & 2's COMPLEMENT

- Let us consider a simple example of 3bit binary system ($r=2, n=3$)



Unsigned Representation:
Only +ve numbers
Range $[0 - (r^n - 1)]$

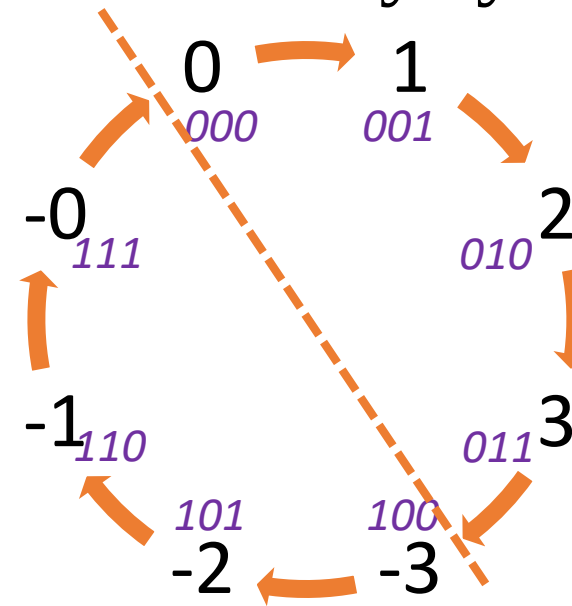


Signed Magnitude:
+ve and -ve numbers

MSB is the Sign Bit:
If $[MSB \text{ eq } 0]$ +ve; else -ve;

Remaining bits represent the
magnitude of the num.

Range $[(r^{n-1} - 1), 0, (r^{n-1} - 1)]$
Distinct values for +ve & -ve 0

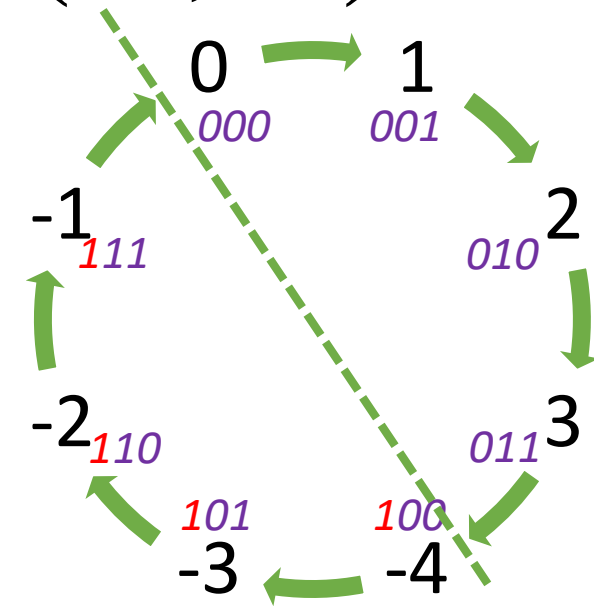


1's complement:
+ve and -ve numbers

MSB is the Sign Bit:
If $[MSB \text{ eq } 0]$ +ve; else -ve;

All bits are used to get the
magnitude of the num.

Range $[(r^{n-1} - 1), 0, (r^{n-1} - 1)]$
Distinct values for +ve & -ve 0



2's complement:
+ve and -ve numbers

If $[MSB \text{ eq } 0]$ +ve; else -ve;

All bits are used to get the
magnitude of the num.

Range $[(r^{n-1}), 0, (r^{n-1} - 1)]$
Unique value for +ve/-ve 0

TWO'S COMPLEMENT NUMBERS

- Don't have same problems as sign/magnitude numbers:

Unsigned Decimal	3 bit Binary Representation	Signed Magnitude		1's complement $-d \text{ or } d' = (r^n - 1) - d$		2's complement $-d \text{ or } d' = (r^n) - d$ [1+ 1's complement]	
		Binary Rep.	Value	Binary Rep.	Value	Binary Rep.	Value
0	000	000	0	000	0	000	0
1	001	001	1	001	1	001	1
2	010	010	2	010	2	010	2
3	011	011	3	011	3	011	3
4	100	100	-0	111	-0	111	-1
5	101	101	-1	110	-1	110	-2
6	110	110	-2	101	-2	101	-3
7	111	111	-3	100	-3	100	-4

Addition doesn't work!
Two representations for 0!

Addition works!
Two representations for 0!

Finally All is well!

TWO'S COMPLEMENT NUMBERS

- Msb has value of -2^{N-1}

$$A = a_{n-1}(-2^{n-1}) + \sum_{i=0}^{n-2} a_i 2^i$$

- Most positive 4-bit number:
- Most negative 4-bit number:
- The most significant bit still indicates the sign (1 = negative, 0 = +ve)
- Range of an N -bit two's comp number:

TWO'S COMPLEMENT NUMBERS

- Msb has value of -2^{N-1}

$$A = a_{n-1}(-2^{n-1}) + \sum_{i=0}^{n-2} a_i 2^i$$

- Most positive 4-bit number: **0111**
- Most negative 4-bit number: **1000**
- The most significant bit still indicates the sign (1 = negative, 0 = +ve)
- Range of an N -bit two's comp number:

$$[-(2^{N-1}), 2^{N-1}-1]$$